PHARMACOLOGY · MCQ PRACTICE

Drug-Induced Liver Injury

SGL 4
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GIT BLOCK

Pharmacology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Questions

1. A pharmacology student asks what DILI stands for and what its most common presenting sign is. What is the answer? EASY

A. Drug-induced liver injury; jaundice is the most common sign
B. Drug-induced liver injury; abdominal pain is the most common sign
C. Direct idiosyncratic liver injury; fever is the most common sign
D. Drug-induced lipid imbalance; pruritus is the most common sign
E. Delayed idiosyncratic liver inflammation; fatigue is the most common sign

2. A student asks how many medications and herbal compounds are known to cause hepatotoxicity, according to the database maintained by the NIDDK. What is the answer? EASY

A. Fewer than 50, catalogued in the FDA Orange Book
B. More than 1,000, catalogued in the LiverTox database
C. Exactly 200, catalogued in the WHO Essential Medicines List
D. More than 10,000, catalogued in PubMed
E. Fewer than 10, catalogued in the BNF

3. A clinician orders liver function tests to distinguish hepatocellular from cholestatic drug injury. Which lab pattern indicates cholestatic injury? EASY

A. Elevated aminotransferases (ALT/AST) only
B. Elevated blood ammonia only
C. Elevated alkaline phosphatase (ALP)
D. Elevated serum creatinine
E. Elevated amylase and lipase

4. A patient is diagnosed with DILI. What is the first step in management? EASY

A. Immediate liver transplantation
B. High-dose corticosteroids regardless of mechanism
C. Lifelong avoidance of all hepatically metabolized drugs
D. Removal/discontinuation of the offending agent
E. Urgent liver biopsy before any other action

5. A researcher is studying racial differences in DILI. Which drug is noted as the predominant agent implicated in DILI among African Americans, versus White Americans? MEDIUM

A. Amoxicillin-clavulanate in African Americans; trimethoprim-sulfamethoxazole in White Americans
B. Isoniazid in African Americans; valproic acid in White Americans
C. Acetaminophen in African Americans; aspirin in White Americans
D. Methotrexate in African Americans; azathioprine in White Americans
E. Trimethoprim-sulfamethoxazole in African Americans; amoxicillin-clavulanate in White Americans

6. A patient develops hepatotoxicity with a brief latent period, a predictable dose-dependent course, marked AST/ALT elevation, and potentially high mortality. Which mechanism of hepatotoxicity does this describe? MEDIUM

A. Intrinsic hepatotoxicity
B. Idiosyncratic reaction
C. Hypersensitivity reaction with rash and fever
D. Nodular regenerative hyperplasia
E. Granulomatous hepatitis from talc exposure

7. A patient develops hepatotoxicity that is unpredictable, has no dose relationship, and has a variable latent period. The reaction is further subdivided into hypersensitivity (immunologic, onset 1-5 weeks, rash/fever/joint pain) and metabolic (aberrant metabolism, onset weeks to months) subtypes. What is this overall mechanism called? MEDIUM

A. Intrinsic hepatotoxicity
B. Idiosyncratic reaction
C. Zonal necrosis
D. Granulomatous injury
E. Chronic cholestasis

8. A 60-year-old man on tuberculosis therapy develops malaise, jaundice, and markedly elevated transaminases after three months of treatment. Isoniazid is identified as the cause. What is the proposed mechanism? MEDIUM

A. Isoniazid directly inhibits bile salt transport proteins, causing cholestasis
B. Isoniazid depletes glutathione, causing NAPQI accumulation
C. Isoniazid is metabolized to mono-acetyl hydrazine, further metabolized by cytochrome P450 to a toxic product causing hepatocellular injury
D. Isoniazid impairs mitochondrial β-oxidation, causing microvesicular steatosis
E. Isoniazid triggers nodular regenerative hyperplasia

9. A 40-year-old woman on long-term tetracycline for acne develops nausea, vomiting, and fatigue, with biopsy showing microvesicular steatosis. What is the proposed mechanism? MEDIUM

A. Tetracycline depletes glutathione stores, causing NAPQI-mediated hepatocellular necrosis in every case
B. Tetracycline competitively inhibits canalicular bile salt transport proteins in every case
C. Tetracycline causes an immune-mediated attack on bile duct epithelium in every case
D. Tetracycline impairs mitochondrial function, decreasing fatty acid oxidation and causing fat accumulation within hepatocytes
E. Tetracycline raises plasma homocysteine, activating hepatic stellate cells in every case

10. A patient taking the NSAID sulindac develops itching and jaundice. What mechanism is proposed for sulindac's hepatotoxicity specifically? MEDIUM

A. Direct depletion of hepatic glutathione stores regardless of dose administered
B. Immune-mediated nodular regenerative hyperplasia regardless of dose administered
C. Impairment of mitochondrial β-oxidation of fatty acids regardless of dose administered
D. Activation of hepatic stellate cells via elevated homocysteine regardless of dose administered
E. Competitive inhibition of canalicular bile salt transport, contributing to cholestatic liver injury

11. A rheumatoid arthritis patient on methotrexate develops liver fibrosis. What is the proposed pathway linking methotrexate to hepatic stellate cell activation and fibrosis? MEDIUM

A. Methotrexate raises plasma homocysteine, generating oxidative stress and activating proinflammatory cytokines that activate hepatic stellate cells
B. Methotrexate depletes glutathione, causing NAPQI-mediated necrosis
C. Methotrexate directly inhibits bile salt export pump proteins
D. Methotrexate causes an IgE-mediated hypersensitivity reaction
E. Methotrexate impairs mitochondrial fatty acid oxidation directly

12. A patient on long-term amiodarone develops hepatomegaly with fatty infiltration on ultrasound; liver function only partially improves after discontinuation. What is the mechanism of amiodarone hepatotoxicity? MEDIUM

A. Amiodarone directly depletes glutathione, causing acute centrilobular necrosis
B. Amiodarone accumulates in the liver, disrupting lipid metabolism and causing phospholipidosis and mitochondrial dysfunction, leading to necrosis and fibrosis over time
C. Amiodarone inhibits canalicular bile salt transport, causing acute cholestasis
D. Amiodarone triggers an acute IgE-mediated hypersensitivity reaction
E. Amiodarone causes granulomatous hepatitis via talc contamination

13. A patient on a sustained-release niacin formulation for hyperlipidemia is noted to have the highest hepatotoxic potential among anti-hyperlipidemic drugs. What pattern of liver injury is typical of anti-hyperlipidemic drugs as a class? MEDIUM

A. Pure cholestatic injury with normal aminotransferases
B. Granulomatous injury indistinguishable from talc exposure
C. Hepatocellular or mixed hepatocellular-cholestatic injury, usually with elevated aminotransferases
D. Zonal necrosis identical to acetaminophen toxicity
E. Nodular regenerative hyperplasia with portal hypertension

14. A patient develops DILI, and the physician notes several patient-level risk factors beyond the drug itself. Which of the following is listed as a risk factor for DILI? MEDIUM

A. Male sex, which is noted as more susceptible than female sex
B. Pediatric age, which is noted as higher-risk than adulthood
C. Low body weight, which is noted as a specific risk factor
D. Genetic polymorphisms in the cytochrome P450 (CYP) isoenzymes
E. Vegetarian diet, which is noted as protective against DILI

15. A 32-year-old bodybuilder using stanozolol develops jaundice with markedly elevated ALP and GGT but only mildly elevated transaminases; bile duct obstruction is ruled out by ultrasound. What is the proposed mechanism, and how is it typically managed? HARD

A. Direct hepatocyte necrosis via NAPQI accumulation, treated with N-acetylcysteine
B. Immune-mediated granuloma formation, treated with high-dose corticosteroids
C. Irreversible nodular regenerative hyperplasia, treated only with liver transplantation
D. Zonal centrilobular necrosis, treated with activated charcoal
E. Impaired bile flow via effects on bile transporter proteins causing intrahepatic cholestasis, usually reversible on discontinuation and treated with ursodeoxycholic acid

16. A 17-year-old on valproic acid for epilepsy develops hepatomegaly, elevated transaminases, and imaging showing fatty infiltration. What dual mechanism is proposed for valproic acid hepatotoxicity? HARD

A. Impaired mitochondrial β-oxidation of fatty acids causing triglyceride accumulation, plus oxidative stress and direct mitochondrial toxicity
B. Competitive inhibition of canalicular bile salt transport alone
C. Depletion of glutathione with NAPQI accumulation alone
D. An IgE-mediated hypersensitivity reaction alone
E. Elevated homocysteine activating hepatic stellate cells alone

17. According to the three-step cascade summarizing DILI mechanisms, which sequence correctly describes how drugs or their metabolites ultimately damage the liver? HARD

A. Impaired mitochondrial function → cell stress via CYP450 oxidation → bile duct obstruction
B. Cell stress (directly or via reactive oxygen species from CYP450-mediated oxidation) → impaired mitochondrial function → triggered immune reactions
C. Triggered immune reactions → cell stress via CYP450 oxidation → impaired mitochondrial function
D. Direct bile salt transport inhibition → cell stress → granuloma formation
E. Hepatic stellate cell activation → cell stress → impaired mitochondrial function

18. On the lecture's DILI classification wheel (grouping drugs by pattern of injury), amoxicillin-clavulanate and valproic acid both appear more than once across categories. In which two categories does amoxicillin-clavulanate specifically appear? HARD

A. Acute hepatocellular injury and granulomas
B. Zonal necrosis and acute cholestasis
C. Chronic hepatocellular injury and chronic cholestasis
D. Steatosis and granulomas
E. Acute cholestasis and zonal necrosis

19. A clinician recalls that both phenytoin and amiodarone are listed as causes of idiosyncratic hepatotoxicity in the lecture's mechanism diagram, but under different idiosyncratic subtypes. Which subtype does each belong to? HARD

A. Phenytoin is listed under the metabolic subtype; amiodarone is listed under the hypersensitivity subtype
B. Both are listed exclusively under the hypersensitivity subtype
C. Both are listed exclusively under the metabolic subtype
D. Phenytoin is listed under the hypersensitivity subtype; amiodarone is listed under the metabolic subtype
E. Phenytoin is listed under intrinsic hepatotoxicity; amiodarone under hypersensitivity

20. A patient on azathioprine for rheumatoid arthritis is found to have nodular regenerative hyperplasia (NRH) on biopsy. What clinical complication is NRH specifically associated with, distinguishing it from typical cirrhosis-related portal hypertension? HARD

A. Fulminant hepatic failure requiring transplantation
B. Hepatocellular carcinoma developing within weeks
C. Reye's syndrome-like microvesicular steatosis
D. Acute cholestasis with markedly elevated GGT
E. Non-cirrhotic portal hypertension

✓ Answer key

Q	Answer	Difficulty	Why
1	**A**	EASY	DILI (drug-induced liver injury) is an acute or chronic hepatic response to a drug or compound; most cases are asymptomatic, but jaundice is the most common sign when present.
2	**B**	EASY	More than 1,000 medications and herbal compounds are known to cause hepatotoxicity, catalogued in the searchable LiverTox database maintained by the NIDDK.
3	**C**	EASY	In cholestatic injury, alkaline phosphatase (ALP) is elevated, whereas hepatocellular injury shows elevation in aminotransferases (ALT/AST).
4	**D**	EASY	Treatment of DILI begins with removal of the offending agent; the prognosis for recovery is usually favorable after discontinuation of the drug.
5	**E**	MEDIUM	The predominant drug implicated in DILI in African Americans is trimethoprim-sulfamethoxazole, while in White Americans it is amoxicillin-clavulanate.
6	**A**	MEDIUM	Intrinsic hepatotoxins cause necrosis of hepatocytes in a dose-dependent, predictable manner with a brief latent period, marked AST/ALT elevation, and potentially high mortality (e.g. acetaminophen).
7	**B**	MEDIUM	Idiosyncratic reactions are unpredictable, dose-independent, with variable latent period, and subdivide into hypersensitivity (immunologic, 1-5 week onset) and metabolic (aberrant metabolism, weeks-to-months onset) subtypes.
8	**C**	MEDIUM	Isoniazid is metabolized to mono-acetyl hydrazine, which is further metabolized by cytochrome P450 to a toxic product causing hepatocellular injury, particularly in patients with genetic susceptibility — an idiosyncratic response.
9	**D**	MEDIUM	Tetracycline is known to impair mitochondrial function, leading to decreased fatty acid oxidation and accumulation of fat within hepatocytes (microvesicular steatosis).
10	**E**	MEDIUM	Sulindac is the NSAID most consistently associated with hepatotoxicity; it competitively inhibits canalicular bile salt transport, contributing to cholestatic liver injury presenting with itching and jaundice.
11	**A**	MEDIUM	Methotrexate therapy raises plasma homocysteine levels, which generate oxidative stress and activate proinflammatory cytokines; these insults activate hepatic stellate cells, leading to liver fibrosis.
12	**B**	MEDIUM	Amiodarone accumulates in the liver and disrupts lipid metabolism, causing phospholipidosis and mitochondrial dysfunction, which can result in hepatocellular necrosis and fibrosis — a chronic process often showing only partial recovery after discontinuation.
13	**C**	MEDIUM	The pattern of liver injury from anti-hyperlipidemic drugs is typically hepatocellular or mixed, with elevation in aminotransferases; the sustained-release formulation of niacin carries the highest hepatotoxic potential, while statins very rarely cause clinically significant liver injury.
14	**D**	MEDIUM	Risk factors for DILI include genetic polymorphisms in the CYP isoenzymes, along with adult age (higher risk than children), female sex (more susceptible than males), and factors such as obesity, diabetes, and chronic viral hepatitis.
15	**E**	HARD	Anabolic steroids such as stanozolol impair bile flow by affecting bile transporter proteins, causing intrahepatic cholestasis; this injury tends to be reversible on discontinuation and is treated with ursodeoxycholic acid (UDCA) to improve bile flow.
16	**A**	HARD	Valproic acid impairs mitochondrial β-oxidation of fatty acids, leading to triglyceride accumulation in hepatocytes, and also induces oxidative stress with direct mitochondrial toxicity, together contributing to steatosis.
17	**B**	HARD	The three-step cascade for DILI mechanisms is: drugs/metabolites cause cell stress directly or through reactive oxygen species generated during CYP450-mediated drug oxidation, which impairs mitochondrial function and triggers immune reactions.
18	**C**	HARD	On the classification wheel, amoxicillin-clavulanate (Augmentin) is listed under both chronic hepatocellular injury and chronic cholestasis, reflecting the mixed/variable injury pattern seen clinically with this drug.

19	**D**	HARD	On the mechanisms diagram, phenytoin is listed as a hypersensitivity example (along with sulfonamides and Augmentin), while amiodarone is listed as a metabolic example (along with isoniazid and ketoconazole) — note phenytoin itself is also listed under metabolic in the diagram, illustrating that some drugs can act through more than one idiosyncratic subtype.
20	**E**	HARD	Nodular regenerative hyperplasia (NRH), a hepatotoxic effect associated with azathioprine, may lead to the development of non-cirrhotic portal hypertension — portal hypertension occurring without the typical architectural distortion of cirrhosis.